

Lab 4: Sampling, Study Design & SLR

STAT 141, Week 4

In this lab, you will

- Practice identifying different sampling strategies
- Conduct exploratory data analysis (EDA) techniques in R
- Build linear regression models with one dependent variable
- Interpret the estimated coefficients of linear regression models
- Use linear regression models to make new predictions

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Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

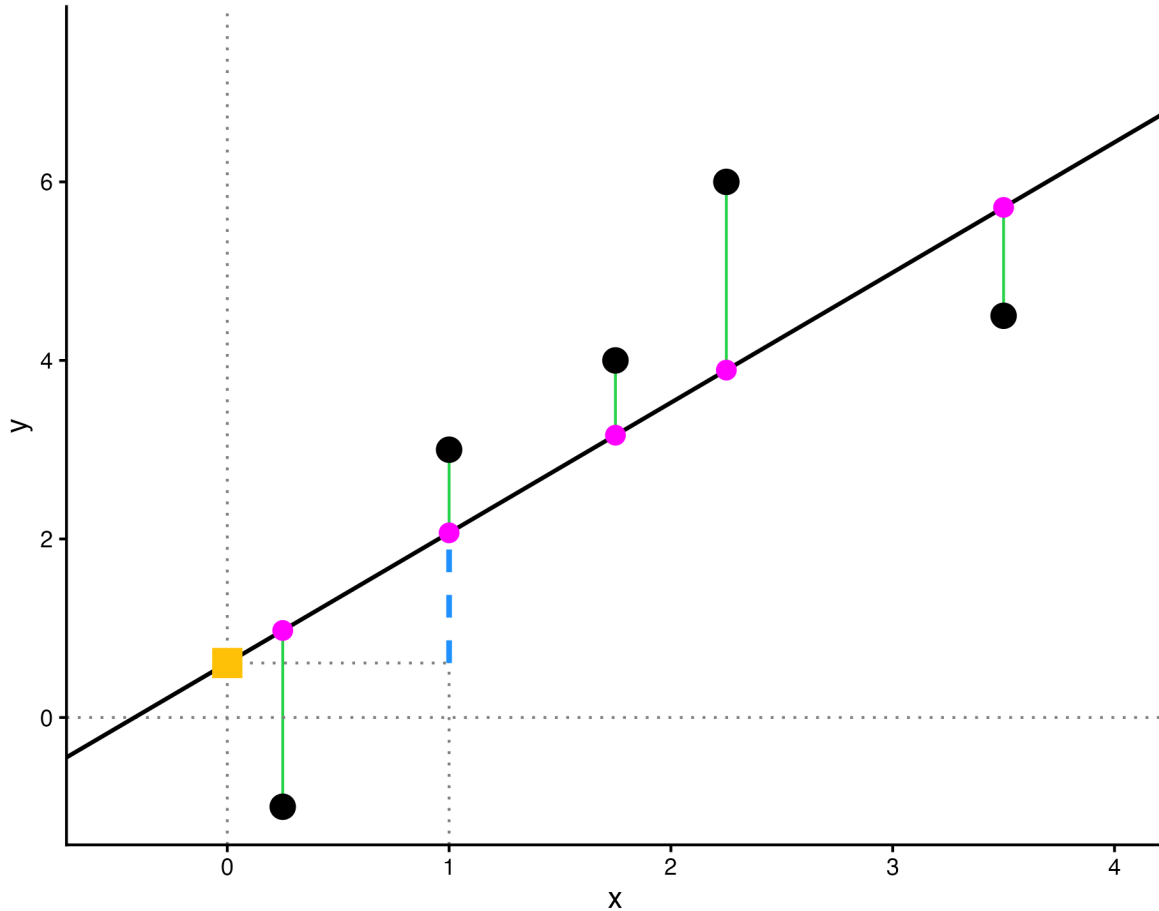
Exercises

Run the following chunk to load the necessary packages.

```
library(tidyverse)
library(pdxTrees)
library(moderndiver)
```

Exercise 1: Anatomy of a Regression

The graph below shows the results of regressing a variable y against a variable x using five data points in black. The graph may be easier to view in the rendered PDF.



Using the vocabulary of regression (fitted values, data values, regression line, residuals, intercept coefficient, slope coefficient), **explain what each of the following elements on the graph correspond to:**

- i. The black solid line
- ii. The blue dotted line
- iii. The green solid vertical lines
- iv. The pink points
- v. The yellow square
- vi. The black points

Use a sentence per element to provide an explanation for what each element represents or measures.

Exercise 2: Regression with Trees Near Reed

This exercise uses data about trees from the parks near Reed. Run the code below to get the data.

```
# Grab trees near Reed
near_reed <- get_pdxTrees_parks() %>%
  filter(Park %in% c("Crystal Springs Rhododendron Garden",
                    "Kenilworth Park",
                    "Eastmoreland Garden",
                    "Berkeley Park"),
         !is.na(Functional_Type))
```

The crown of a tree refers to its total above-ground parts, including stems, branches, and leaves. In the `near_reed` data frame, `Crown_Width_EW` represents the greatest width of the crown in the east-west direction, while `Crown_Width_NS` represents the greatest width of the crown in the north-south direction. The code below adds a new variable to the `near_reed` data frame called `Crown_Width` which is the average of `Crown_Width_EW` and `Crown_Width_NS`.

```
near_reed <- near_reed %>%
  mutate(Crown_Width = (Crown_Width_EW + Crown_Width_NS) / 2 )
```

a. Generate a scatterplot with `Crown_Width` as the explanatory variable and `Tree_Height` as the response. Be sure to include an informative title and axis labels to your plot using `labs()`. [Code only for this question]

b. Do you expect the correlation coefficient (for `Crown_Width` and `Tree_Height`) to be positive or negative? Write a code chunk testing your prediction. [Code and text for this question]

We'll see these more in lecture tomorrow, but let's preview two essential functions in R for working with linear regression models. The first function is `lm()`, which takes in a model equation and our data and returns a fitted linear model. For example, we might run something like: `mod <- lm(y ~ x, our_df)` to fit a linear model that uses `x` to predict `y`, where `x` and `y` are both columns in the data frame `our_df`.

The direct output of `lm()` is not super informative, so we'll usually just save it to an object to use later (`mod` above). The second function that we'll use all of the time when working with linear models is `get_regression_table()`. This function is from the `moderndive` package, and allows us to see some nicely formatted results about our model, including coefficients. It takes in a model created by `lm()` and returns a nice table. For example, we would run `get_regression_table(mod)` for our above model.

c. Using the `near_reed` data frame and these two new functions, run a linear regression using `Crown_Width` as the explanatory variable and `Tree_Height` as the outcome variable, and display a table of coefficients. [Code only for this question]

d. Carefully interpret the estimated coefficients for the intercept and the slope term for Crown Width, *in context*. These can be found under the “estimate” column in the output of `get_regression_table()`. For now, we'll ignore the other columns. [Text only for this question]

e. Write out the equation for the regression line, aka the “fitted model”, using these coefficient estimates.

f. Make a numerical prediction for the `Tree_Height` for a tree that has 40 foot crown width, both using the formula for the linear model you created above **and** by using the `predict()` function. [Code is needed for this question, and you should write your final answer out in text.]

Exercise 3: Voter ID Laws and Sampling

Back in 2014, researchers investigated the impact of voter ID laws on potential voters in the midterm elections. The study involved three locations: Philadelphia, Northern Virginia, and Dallas County, Texas. Each location is divided up into voting precincts and each precinct has a polling site.

In this problem, we will describe the data collection method of two locations and you will practice tying these methods to specific sampling techniques.

a. For Philadelphia, 30 precincts were randomly selected from all precincts in Philadelphia. For each selected precinct, Swarthmore College students interviewed every 3rd person leaving the polling site over a randomly selected hour. **Does this sampling design use cluster sampling, stratified sampling, neither, or both? Justify your answer.**

b. For Dallas County, each polling site was categorized by the time of day interviews would take place (morning/mid-day/evening) and whether voters at that site were projected to have an overall high or low propensity of voting. Within each category, a random sample of precincts were selected and Southern Methodist University students interviewed every person leaving those polling places at the particular time of day (morning/mid-day/evening). **Does this sampling design use cluster sampling, stratified sampling, neither, or both? Justify your answer.**

c. Why didn't the researchers just take a simple random sample of names (i.e., from a voter registration list) and then conduct exit polls on those people? Justify your answer.

d. Suppose that for a particular state, precincts are known to be relatively heterogeneous; e.g., precincts tend to have different social demographics. **Would cluster sampling or stratified sampling be the recommended strategy for this state? Justify your answer.**
